1 Fig. 1.1 shows a small solid metal cylinder of mass m, length L and diameter d.

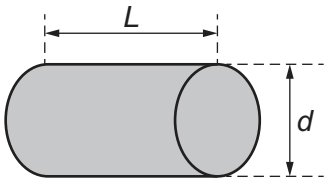


Fig. 1.1

The cylinder is heated to a uniform temperature. The cylinder is then removed from the heat source and the cylinder is wrapped in an insulating material.

The temperature of the room is T_R. At time t after the cylinder starts to cool, the surface temperature of the cylinder is T_C.

It is suggested that T_C is related to t by the relationship

$$(T_C - T_R) = Ze^{-\frac{UAt}{mc}}$$

where A is the total surface area of the cylinder, c is the specific heat capacity of the metal, and U and Z are constants.

Plan a laboratory experiment to test the relationship between T_C and t.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for U and Z.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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